

CHARACTERISATION AND OPTIMISATION OF 3D-LCD MODULE DESIGN

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1. Abstract

Autostereoscopic displays with flat panel liquid crystal display and lenticular sheets are receiving much attention. Multiview 3D-LCD is truly autostereoscopic because no head tracking is necessary and the technology is well poised to become a mass market consumer 3D display medium as the price of liquid crystal displays continues to drop. Making the viewing experience as natural as possible is of prime importance. The main challenges are to reduce the picket fence effect of the black mask and to try to get away with as few perspective views as possible.

Our solution is to 'blur' the boundaries between the views. This hides the black mask image by spreading it out and softens the transition between one view and the next, encouraging the user to perceive 'solid objects' instead of a succession of flipping views. One way to achieve this is by introducing a new pixel design in which the pixels are slanted with respect to the column direction. Another way is to place the lenticular at a small (9.46°) angle with respect to the LCD columns. The effect of either method is that, as the observer moves sideways in front of the display, he always 'sees' a constant amount of black mask. This renders the black mask, in effect, invisible and eliminates the picket fence effect.

2. Introduction

Active Matrix Liquid Crystal Displays and lenticular sheets provide a straightforward and optically efficient way of making high quality 3D displays^{1,2}. In a previous paper we have presented some 3D-LCD prototype designs that were suitable for individual or professional use³. In this paper we consider how the current generation of SVGA colour LCDs can be used to create a 3D display for general use/

Multiview, lenticular 3D-LCD is well poised to become a mass market 3D display for a number of reasons. First of all it provides a high quality image. Secondly, the price of 3D-LCD is ultimately dominated by that of the LCD itself. The price of these LCDs is expected to continue the aggressive erosion of the past years, particular in the large diagonal sizes. For instance, by the year 2000, AM-LCDs equivalent in size to 21" CRT monitors are expected to be equal in price⁴. A further, third reason in favour of 3D-LCD is that the combination of lenticular and display is both simple and forgiving, keeping down the additional cost of making an LCD into a 3D display. A final, fourth reason for 3D-LCD forms the focus of this paper, namely that from a user perspective, 3D-LCD provides a very *natural* 3D experience.

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Traditional 3D display design emphasises the requirements of professional use, with much attention to the correctness and quality of the stereo image. These requirements are often met by introducing complicated and obtrusive technology, which, paradoxically, and in spite of the high quality stereo image, stands in the way of any intimacy between the user and the visual objects. The effect is, that although the user does see a high fidelity 3D image, he is also continuously aware of the technology that is providing him with that image. HMDs are an obvious example of this because it is very difficult to forget that you've got one on your head. Autostereoscopic systems with headtracking also have this drawback, albeit in a much smaller degree. It is however unnatural that the display system has to 'know' that you're looking at it and that at any time only a limited number of people can look at the display and get the stereo benefit. For the mass market, discounting the temporary novelty factor, naturalness and ease of use are, after cost, the most important parameters. Hence, if there are design trade-offs between the 'naturalness' of the 3D experience and 'correctness' of the stereo image, mass market consumer applications will favour the former.

This perspective is helpful for 3D-LCD, where the main technical challenges are to reduce the picket fence effect of the black mask and to try to get away with as few perspective views as possible. These problems are related since the black mask image is the boundary between views and a limited number of views makes these boundaries obvious, even if the width of the black mask could be reduced to a minimum. The solution offered by a trade-off between correctness and naturalness is to 'blur' the boundaries between the views. This will hide the black mask by spreading it out and soften the transition between one view and the next, encouraging the user to perceive 'solid objects' instead of a succession of flipping views.

This paper investigates this issue by first considering in detail the effect of the very regularly ordered pixel structure of LCD. It then considers the effect of a random pixel structure, which, although technically unfeasible is conceptually helpful. The paper then discusses two practical ways in which some of these concepts can be realised. One way is to give the LCD pixels a tilted parallelogram shape and the other is to slant the cylindrical lenses at a small angle with respect to the LCD column direction. A practical implementation of the latter is discussed.

3. LCD Pixelisation

LCDs are very suited to lenticular imaging because they are flat, have thin face plates and have very accurately defined pixels. This last feature allows the precise identification of perspective views with pixels on the LCD. It has however also a down side. As illustrated in Figure 1, the perspective views are simply projections of the discrete LCD pixels. This has two effects, first of all the pixel boundaries are also imaged and form an array of vertical bars through which the user has to look at the display. The visual effect of this is that, with sideways movement, Moiré-like fringes appear to run over the screen. A second effect of the discrete views is that the image on the eyes remains constant when the user moves sideways within one pixel projection. This is unnatural for a 'real' 3D object and the consequent distortion of the stereo image has been considered by a number of people^{5, 6}. The visual effect is an apparent rotation of the object which, in multiview systems, periodically 'flips' back. Figure 2 illustrates this effect by plotting what we call the view angle error $\Delta\phi$, which is the difference between the angle at which the user is looking at (the centre) of the display and the angle under which the camera took the original picture. Obviously the former varies continuously, whereas the latter only varies in discrete steps, generating the saw-tooth difference marked 'Standard LCD' in Figure 2. Whether these effects of the discrete viewing zones are acceptable artefacts or a barrier to user acceptance, depends on the application. In the following sections we discuss how, for applications that can afford to make the trade-off between 'correctness' and 'naturalness' the 3D-LCD design can be optimised to mitigate these effects.

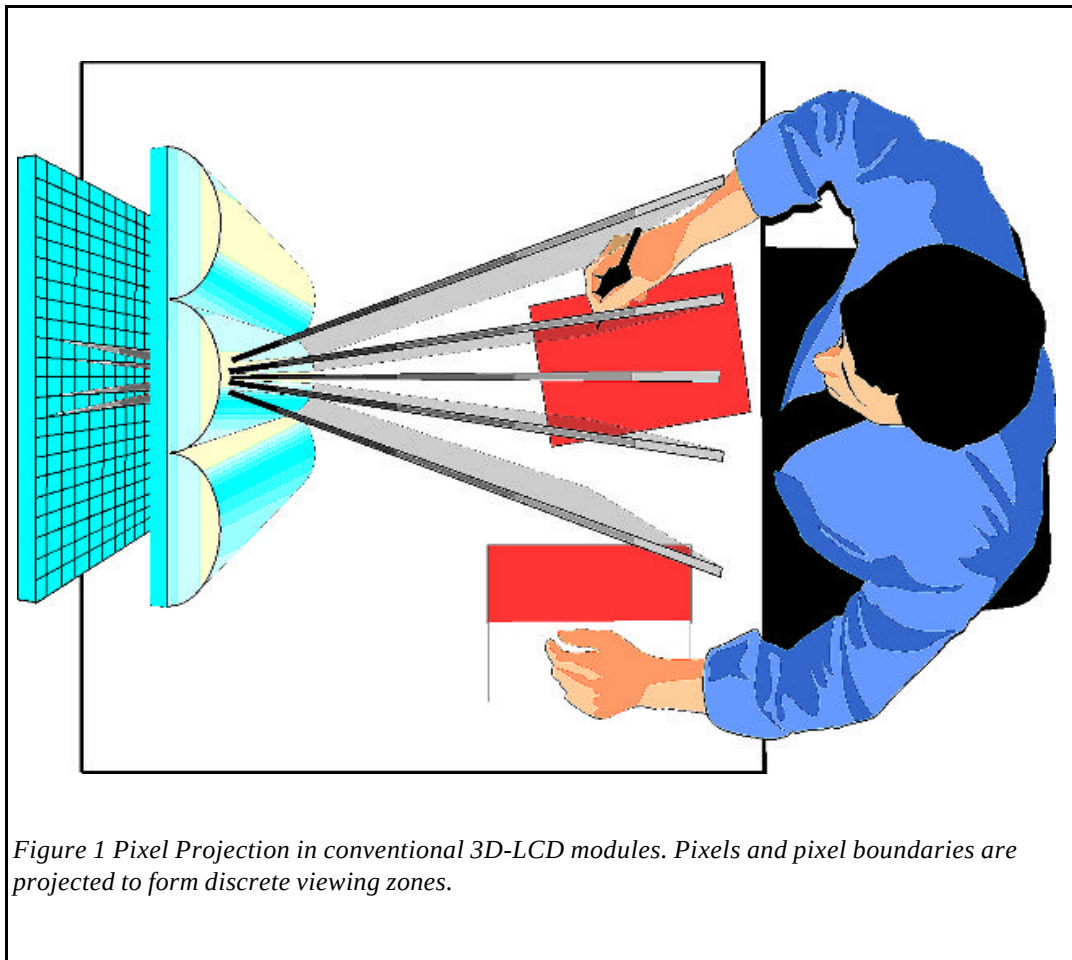
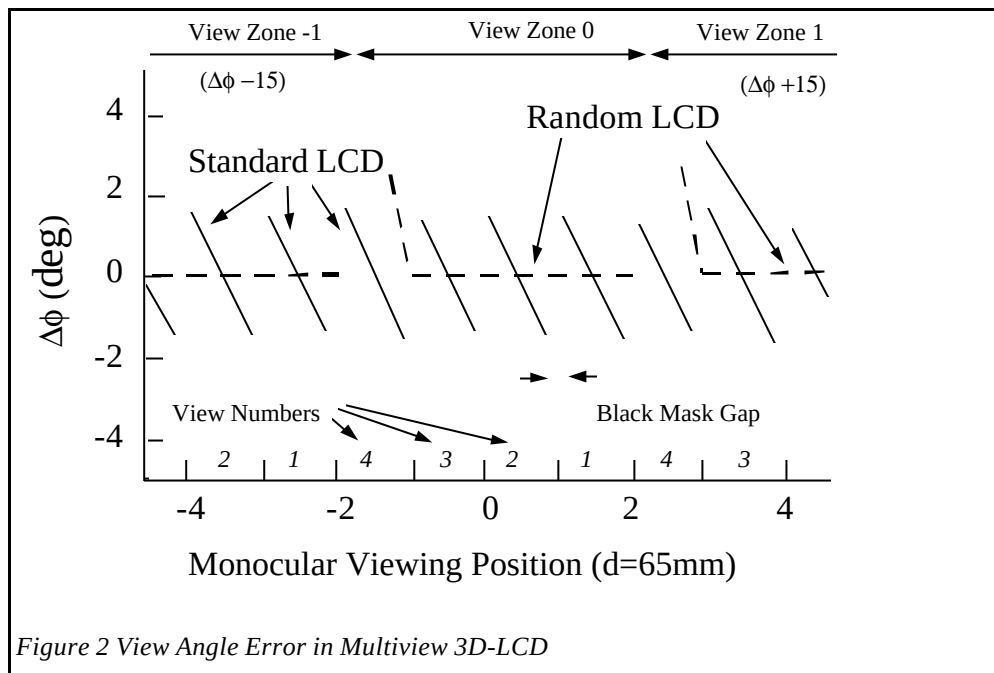


Figure 1 Pixel Projection in conventional 3D-LCD modules. Pixels and pixel boundaries are projected to form discrete viewing zones.

4. Random Pixel LCD

A very different situation exists in 3D photography and postcards. Here the image carrying medium is not a pixelated LCD but a photographic print with a random distribution of dye dots. There are no precisely defined views because the size of each individual dye dot is much larger than the width of pixels would be in an equivalent 3D-LCD. I.e. the pitch of the lenticular lens divided by the number of views. Yet the visual impression is of a continuous parallax picture, even if the depth range is limited and planes outside the datum plane are a little out of focus. The key lies in the placement of the dye dots that is in *detail random*, but on *average accurate*.

We could carry this idea to LCD and propose an LCD with a random pixel distribution. Such a suggestion may be technically unworkable but is conceptually interesting. Suppose we had an LCD that is standard in every way except that each row is off-set from a reference position by a random amount. Suppose also that we know that amount for each row, that we can feed that information back to our image generation system (cameras, render engine) and supply each row with the perspective views that correspond to the centre of the random position of the pixels. This may sound convoluted, but is of course exactly what happens to the photographic crystals in single lens lenticular integral imaging^{7,8}. In our random pixel LCD, the display information density is the same as for a standard LCD because the total number of pixels has not changed. Now the saw-tooth in Figure 2 is no longer representative of the whole display, but



only of one row. The saw-tooth of each individual row will have a random offset and to get a representation of the display as a whole we must take an average over a reasonable number of rows. As might be expected, the random saw-tooths cancel each other out and the plot for the whole display follows the dashed line in Figure 2. This *suggests* that the error is zero and that pseudo movement and flipping have been eliminated but that is strictly speaking not the case; flipping still takes place, but now occurs on a row-by-row basis instead of over the whole display simultaneously. This makes it less obvious and in practice it means that the subjective impression of flipping *has* been eliminated.

More important is that, in the random pixel LCD, the image of the black mask has disappeared. In Figure 2 the saw-tooth labelled 'Standard LCD' is discontinuous because in the areas where the black mask is imaged the error $\Delta\phi$ is not defined. In the random pixel LCD however, these gaps cancel each other out and the dashed line labelled 'Random LCD' is continuous within each viewing zone, indicating that along with the regular pixel structure, the black mask image has disappeared.

The price that has been paid for this is that there is now crosstalk between the views. Interocular cross talk between views reaching the left and right eye can limit the fuseability of the 3D scene⁹. On the other hand, it has been recognised that in a multiview system, crosstalk can help to reduce the flipping between views¹⁰, reducing the number of views required, or conversely increasing the apparent number of views.

5. Slanting Pixels

Figure 3 illustrates a novel pixel design for a 4 view system, that eliminates the image of the pixel gaps in the 3D-LCD. As can be seen, instead of the usual rectangular pixels, the pixels are skewed or slanted and have the shape of a parallelogram. The area between the blocks represents the LCD black mask. In discussing the figure, it is important to realise that, as illustrated in Figure 4, because the LC cell is located in the focal plane of the cylindrical lens, horizontal position x in the LCD corresponds to viewing angle f . Hence all points on the dashed line A are seen simultaneously by one eye of the user at one specific horizontal viewing angle, as are all points on B from a different viewing angle. Line A represents a (monocular) viewing position in which only pixels belonging to view 2 can be seen. Line B represents a

(monocular) viewing position in which pixels from both views 2 and 3 can be seen together. Line C in turn represents a position in which only pixels from view 3 can be seen. Thus as the observer moves with one eye closed between positions corresponding to line A and C, he will see a changeover in which view 2 fades out and view 3 simultaneously fades in. If the slant angle of the pixels is such that the bottom left hand corner of each pixel aligns vertically with the top left corner of the pixel in the next column, then the amount of black mask along every vertical line is the same. Hence the observer 'sees' an equal amount of black mask from every position and this uniform visibility makes it in fact *invisible*.

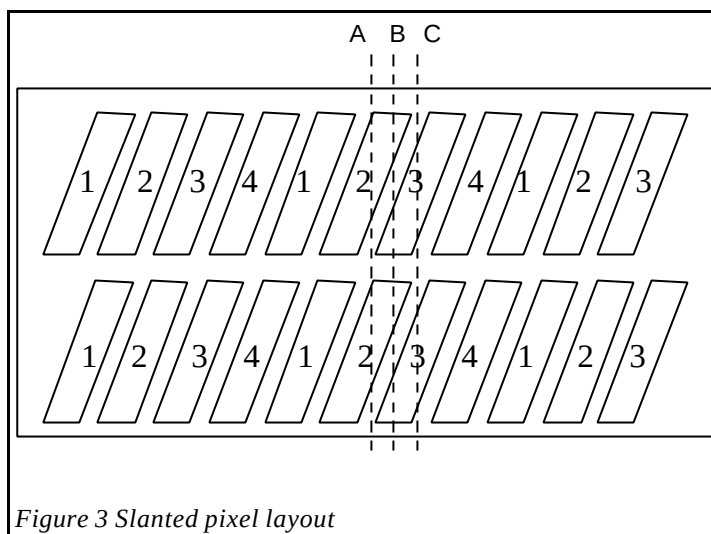


Figure 3 Slanted pixel layout

Provided the disparity between the views is not too great, the fading between the pixels of adjacent views will encourage the user to interpolate between the views and in this way perceive more views than there actually are¹⁰. We have confirmed this using a dummy LCD 'fabricated' with a high resolution laser printer¹¹. As expected, no black image of the black mask can be seen. Furthermore, using static pictures on the dummy LCD we were able to show good 3D pictures.

Other pixel shapes, such as triangles or hexagons, can also be used to achieve similar results. In this respect it is interesting to note that a very similar effect to that of the slanted pixels was achieved in a projection system through the use of hexagonal aperture stops¹².

Although perfectly regular, the slanted pixel arrangement has some commonality with the random pixel LCD discussed above. The effect of the random offsets of the rows of the random pixel LCD is achieved here within each row by the shape of the pixel, in which each horizontal slice of a pixel can be said to be offset from the slice below. The

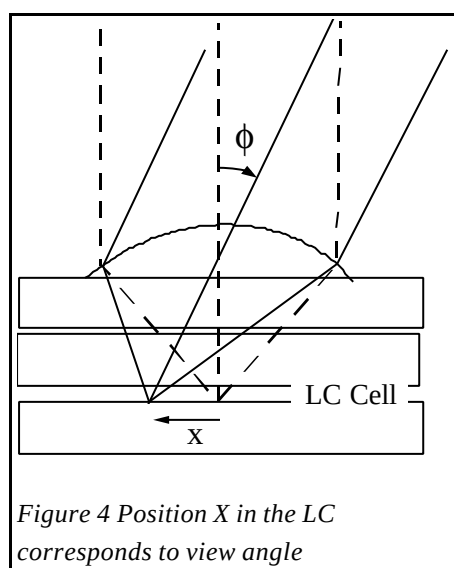


Figure 4 Position X in the LC corresponds to view angle

'average over a reasonable number of rows' in the discussion of the dashed line in Figure 2 now becomes an integral over the shape of the pixel parallelogram. An important difference however is that each horizontal pixel slice does not carry the correct information corresponding to the centre of its position in the way that each offset pixel did in the random pixel LCD. Instead, each pixel carries the same information (that of view 1, 2 etc) on all its slices. This means that along a given vertical line, the view angle error $\Delta\phi$ is constant and the saw-tooths for the slices are not just offset horizontally, but also vertically in such a way that they all form segments of a single line. The averaging now results in a line not at zero, but at a small negative value. The fact that this value is constant with view position x can be seen more easily from Figure 3. The view angle error along line B will have a contribution from the error in view 2 and the error in view 3. When the user then moves a small amount towards the position corresponding to line C, the proportion of error from view 3 increases, but the value of that error

decreases because the movement is towards the centre of the pixel where the error is zero. Similarly the contribution from view 2 decreases while the error increases because the movement is away from the centre of view 2. It is intuitively obvious that the total view angle error $\Delta\phi$ remains constant.

6. Slanting Lenticulars

The same effect can be achieved in a different way by using a conventional, straight pixel, LCD but putting the lenticular sheet at a slant. This is illustrated in Figure 5 for the case of a 7 view system. The height to width aspect ratio of standard LCD pixels is 3 to 1 and the lenticular is aligned at an angle of $\arctan(1/6) = 9.46^\circ$ to the LCD column direction.

The dashed lines in Figure 5 again help us to see how the pixels in the LCD group themselves into views. Line 'A' corresponds to a position in which view 3 only can be seen. Under each lenticular, the pixels of view 3 are positioned on alternate row lines. At position A, the user sees an image of the black mask on the interspersed row lines. When, as before, the user moves his head sideways, the imaging optics of the cylindrical lens picks up the pixels on the interspersed row lines and view 4 comes into view. The transition between the views is gradual and an identical fading of the views as in the slanted pixel arrangement above, is achieved. The black lines have disappeared and the fading achieves an effect of a solid object rather than flipping views.

Slanting the lenticular in this way emulates the effects of the random pixel LCD discussed above. If we look along the cylindrical lens we see that the odd rows have an offset of a pixel pitch with respect to the even ones. Giving the pixels on odd and even rows their own view number means that, as in the random pixel LCD, each pixel is part of the perspective view that corresponds to (the centre) of its position

Any number of views can be achieved in this way. The 7 view example is interesting because it shows a distribution of views which is offset by one row underneath adjacent cylindrical lenses and illustrates the versatility of the scheme.

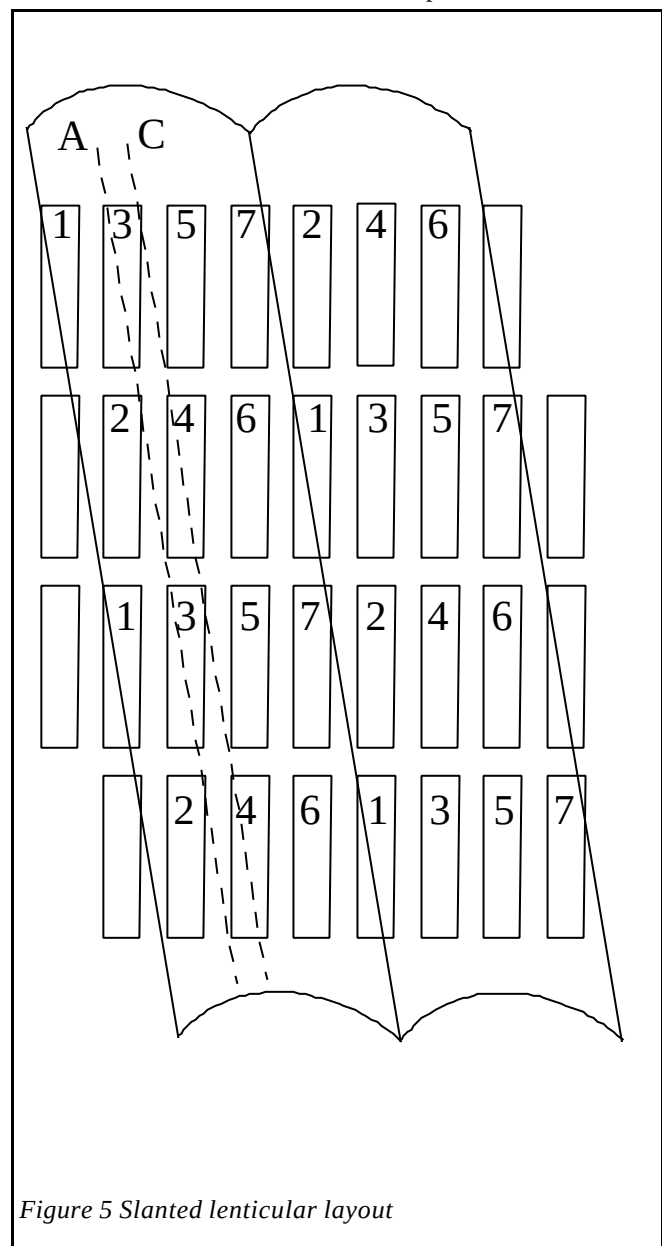


Figure 5 Slanted lenticular layout

7. 7 View 3D-LCD

We have built a prototype display using this idea. The LCD used was an SVGA panel from Flat Panel Display Co in Eindhoven, the Netherlands and the lenticular sheet was manufactured by Philips Optics, also in Eindhoven. The details of the 3D-LCD are summarised in table 1.

LCD	Resolution (Pixel Count)	800(x3)x600
	Size (Diagonal)	11.4"
	(Sub) Pixel Pitch (μm)	96 x 288
	Colours	262144
Lenticular Sheet	Size (mm)	260 x 210
	Pitch(μm)	331
	Focal Length (mm)	0.99
3D-LCD	Number of Views	7
	Resolution / View	342 x 200 x RGB
	Working Distance (mm)	300 - 1500

Table 1 Specification of 7 view 3D-LCD

In the design of the prototype the magnification was chosen such that at the specified working distance, next nearest neighbour views, for instance views 2 and 4, are projected into the right and left eyes respectively. This means that the crosstalk between views inherent in the slanted lenticular design, does not become inter-ocular crosstalk. Using an Interocular distance of 65mm, this then gives a magnification of 677. The specified pitch of the cylindrical lenses is measured perpendicular to their long axis and is a factor of $\cos(9.46^\circ)$ less than the required pitch along the row direction of the LCD. The working distance of the display is limited by the minimum distance between the LC plane and the lenticular sheet. The lenses were fabricated on a separate substrate which was then placed face down on the LCD. This achieves a minimum separation between LC cell and lenses and results in a working distance of 67 cm. This working distance is however nominal only. Because there is no black mask image and no sharp boundaries between views, the display can be looked at from a wide range of distances ranging from 0.3m to 1.5 meters.

The number of RGB colour triplets in each of the 7 views in the 3D-LCD is 342 x 200. This raises one of the key advantages of the slanted lenticular scheme, namely that both the horizontal and vertical pixel count in the LCD is used to trade-off against the number of views. In a conventional design, only the horizontal pixel count can be used. This would mean that for a colour 7 view system based on an SVGA display, the resolution per view would be 114 (h)(x3) x 600. This gross imbalance between horizontal and vertical resolution would clearly be unacceptable. The slanted lenticular scheme however achieves a much better balance between horizontal and vertical resolution.

The layout of the pixels as they appear to the eye in a view is illustrated in Figure 6. As can be seen the colour triplets are arranged diagonally and it is this that lies at the heart of the resolution balance. Figure 6 also illustrates a further advantage of the slanted lenticular scheme. The figure illustrates not only the pixels of view 4, but also the pixels of view 3 as indicated in the greyed-out boxes. As can be seen, the pixels of both views are located in different spaces. This means that, when the observer is looking from an angle under which he sees both views due to the inherent crosstalk, and if these views contain correctly sampled pixels of an identical image, he will see that image at a *higher* resolution than each of the views individually allows. This feature is significant because for objects at the datum plane, the image in each view is identical and it

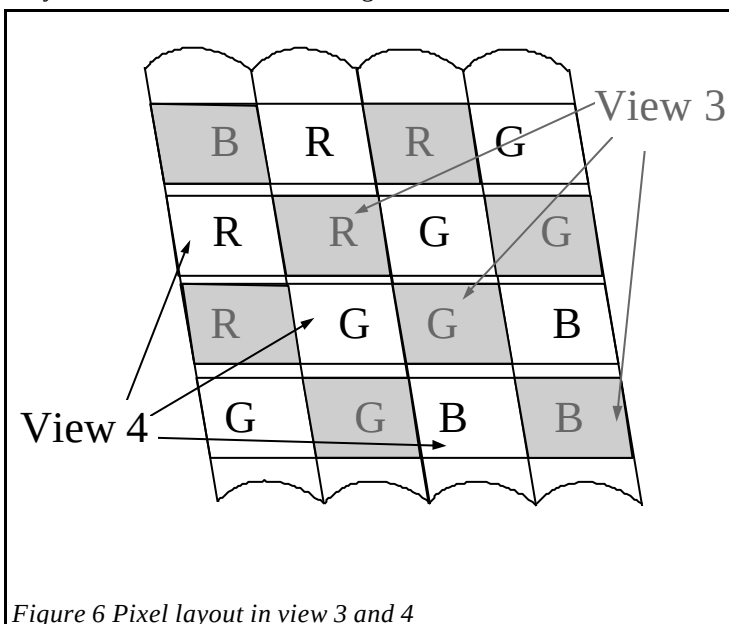


Figure 6 Pixel layout in view 3 and 4

therefore follows that the slanted lenticular arrangement allows the representation of objects at the datum plane at a higher resolution. For objects outside the datum plane the crosstalk between the views causes a blurring which is not unlike the blurring others^{13,14} have introduced intentionally into stereo images to create comfortable depth perception.

8. Conclusions

In this paper we have considered the LCD pixelisation as a key characteristic in the 3D-LCD performance. Through analysing the effect of the pixelisation and by speculating about the effect of a random pixel distribution we have arrived at two design options that optimise the 3D-LCD for applications where an overall 'naturalness' of the image is more important than the 'correctness' of the stereo image in detail. The most powerful of these two options is to place the cylindrical lenses of the lenticular sheet not parallel to the LCD column direction, but at a small angle to it. This provides a number of benefits. Chiefly these are a fading of the clear boundaries between views that would otherwise exist and a balance between horizontal and vertical resolution in each view that does otherwise not exist. Specifically these benefits result in the elimination of the black mask picket fence effect, an increased freedom of head movement, particularly viewing distance, a resolution for objects at the datum plane that exceeds the resolution of each individual view, an automatic blurring outside the datum plane, the use of off the shelf LCD modules and a high view count. It is expected that the combined effects of these benefits will make 3D-LCD an attractive 3D display medium for general, mass consumer use.

9. References

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